Samuel Louviot, Ph.D

SCIENTIFIC SOFTWARE ENGINEER

ABOUT ME

Neuroscientist and engineer with a diverse work experience, seeking a software engineering position to utilize a broad skill set in engineering and neurophysiology. A technology enthusiast committed to continuous learning and fostering cutting-edge technology for healthcare and environmental preservation.

PERSONAL DETAILS

- Onsite | Hybrid | Remote
- samuel.louviot@pm.me
- (716) 203-5107
- github.com/Sam54000
- in/samuel-louviot/

EXPERIENCE

Washington University | NCAN

SCIENTIFIC CONSULTANT (VOLUNTEER) Jan 2022 - Present

Consulting on transcranial electrical stimulation research in Humans with invasive EEG

Nathan Kline Institute | Child Mind Institute

SCIENTIFIC SOFTWARE ENGINEER GENERALIST | EEG EXPERT Jan 2024 - Present

• Researched, developed and deployed machine learning models to predict fMRI brain state from electroencephalograms (EEG) during naturalistic stimuli, achieving a maximum of 70% prediction accuracy.
• Developed software pipelines to handle multimodal data (EEG, fMRI, eyetracking) and design data model to streamline Machine Learning models training.
• Designed and developed EEG data cleaning and quality assessment pipelines for a cohort of 260 human subjects on Linux server.
• Developed Python package dedicated to EEG research.
• Mentored data analysts in Neural Engineering. Mentored students in data analysis and machine learning techniques.

Weill Cornell Medicine

POSTDOCTORAL ASSOCIATE | DATA SCIENTIST | NEURAL ENGINEER Jun 2022 - Jan 2024

• Designed and developed EEG analysis software systems and data pipelines in Python.
• Researched and developed machine learning models dedicated to Brain Computer Interfaces (BCI) reaching more than 60% accuracy.
• Conducted Big Data analytics on multi-modal data in a cohort of over 140 participants.
• Collaborated with engineering teams in industry and academia to develop hardware and software solutions for high density closed-loop transcranial electrical stimulation in Brain Computer Interfaces (BCI)
• Mentored students in research methods, data analysis, and software engineering

Hospital Central, Université de Lorraine, CNRS

GRADUATE RESEARCH ASSOCIATE Oct 2018 - Jun 2022

• Data analytics of biophysical models of intracerebral electric field in Matlab and Python
• Mentored undergraduate students in data analysis and research techniques

Université de Lorraine

LECTURER 2020 - 2021

Taught the course Applied Mathematics for Biomedical Engineering at the Graduate School of Biomedical Engineering.

EDUCATION

Université de Lorraine, CNRS

PH.D, NEUROSCIENCE | NEURAL ENGINEERING 2022

Université de Lorraine

MS, BIOMEDICAL ENGINEERING, CYBERNETICS, ROBOTICS, AND IMAGING. 2018

Université de Lorraine

BS, BIOMEDICAL ENGINEERING 2016

Lycée Pierre Mendes France

ASSOCIATE DEGREE, ELECTRICAL ENGINEERING 2014

SKILLS

Machine Learning	Deep Learning	Python
Bash	SQL	System Design
Scikit-learn	Linux	Windows
macOS	TensorFlow	PyTorch
Matlab	Brain Computer Interfaces	Electrical Engineering
C++	Rust	Microcontrollers
Experiment Design	CAD	Signal Processing
AWS	IBM Watson	Cloud Computing
Model Deployment		

LANGUAGES

French	English	Spanish